# Precise calcium-to-spike inference using biophysical generative models

This Figshare repository is in support of "Precise calcium-to-spike inference using biophysical generative models" from Broussard et al. It contains the data exports, MATLAB plotting functions, Python plotting scripts, shared utilities, and table-generation inputs used to reproduce the manuscript figures, supplementary figures, and tables.

Figure panels are generated by scripts in the matching figure* or supp_figure* directory. MATLAB panels are generally script-style functions named for the figure and panel range. Python panels are installable console entry points defined in setup.py.

## Environment and paths

The revision version of record was tested with MATLAB R2023b Update 6 and Python 3.12.2. MATLAB scripts use functions from the Curve Fitting, Optimization, Signal Processing, Statistics and Machine Learning, and System Identification toolboxes. The exact Python package versions are pinned in requirements.txt and setup.py.

From the repository root on a Windows machine:

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
pip install -e . --no-deps
```

Conda works equivalently after activating the target environment. On offline or restricted HPC nodes, install setuptools<82 and wheel into the environment first, install requirements.txt, and then run pip install -e . --no-deps --no-build-isolation. Python entry points resolve data and output paths back to this project checkout, not to the environment's site-packages directory. If launching from an unusual working directory, set GCAMP_FIGSHARE_ROOT or FIGSHARE_PROJECT_DIR to the repository root.

For MATLAB, add the full checkout to the path before running figure scripts:

```
repoRoot = 'C:\path\to\Figshare';
addpath(genpath(repoRoot));
```

After that, run scripts from the repository root or from their figure directory. Generated PNG, vector PDF, editable MATLAB FIG, statistics, and table-rendering artifacts are written beneath the repository-level output/<figure-or-table> directory, which is intentionally excluded from Git.

## Main figure contents

- figure1: jGCaMP response properties. Panels cover single-spike responses, response amplitude/kinetic summaries, paired-pulse and multi-spike response behavior, and jGCaMP8f in vivo quiescent-event examples with last-aligned averages and fast decay summaries.
- figure2: stopped-flow and bouton kinetic measurements across GCaMP variants. Scripts plot on/off kinetics, stopped-flow rise times, bouton rise comparisons, Hill-style summaries, 10-pulse responses, decay summaries, and kinetic contribution analyses.
- figure3: biophysical generative model fits and spike-inference examples. MATLAB scripts show stopped-flow and bouton/state fits for GCaMP6f, jGCaMP7f, and jGCaMP8f. The Python script compares biophysical and linear SMC inference on simulated excitatory/inhibitory examples.
- figure4: spike-inference performance on jGCaMP8f data. MATLAB scripts plot example continuous/discrete spike predictions and correlation versus smoothing width for excitatory and inhibitory cells using correlation_tbl_jG8f.mat. The Python script plots matched bias/recall-style distributions.

## Supplementary figure contents

- supp_figure1: saturation, elongation, and related controls for jGCaMP8f multi-spike protocols.
- supp_figure2: EGTA/BAPTA clamp comparisons, including data and simulations for OGB and jGCaMP8f.
- supp_figure3: magnesium and ionic-strength effects on steady-state fluorescence and kinetics across sensors.
- supp_figure4: stopped-flow decay traces, biexponential fits, and decay-component summaries.

- supp_figure5: clustered parameter-set summaries used during GCaMP model parameter selection.
- supp_figure6: model fits and model-state plots using the parameter JSON files stored with Table 3.
- supp_figure8: explicit-buffer simulations and Biophys SMC inference on simulated pyramidal-like and interneuron-like cells. Run with run_supp_figure8_code; the script uses compact processed CSV exports for the plotted trajectories and posterior spike summaries.
- supp_figure9: legacy/additional jGCaMP8f correlation analyses. Current Figure 4 analyses use figure4/correlation_tbl_jG8f.mat.
- supp_figure10: example high/low SNR spike predictions and discrete spike timing metrics.
- supp_figure11: spike-history effects on false-positive and false-negative inference errors for Biophys SMC, MLspike, and PGBAR. suppfigure11_code.m uses the packaged plotting summaries; suppfigure11_stats.m reruns the panel B/C generalized mixed-effects analyses from packaged cell-epoch contributions.
- supp_figure12: STA/SNA bias and uncertainty distributions for matched or unmatched jGCaMP8f detections. Run with run_supp_figure12_A_code.
- supp_figure13: trialwise jGCaMP8f analyses using packaged CSV/NPZ exports. Run with run_supp_figure13_code; the script writes the raincloud panel and combined trace panels only.
- supp_figure14: jGCaMP8f Janelia-parameter comparison. suppfigure14_A_B_code.m uses janelia_model_selection_data_jG8f.mat; run_supp_figure14_C_D_code plots trace and correlation-vs-sigma comparisons.
- supp_figure15: jGCaMP8m analyses. suppfigure15_A_B_C_code.m uses decay_v_spikeNum.mat for quiescent binning; suppfigure15_D_E_code.m uses janelia_model_selection_data_jG8m.mat; run_supp_figure15_F_G_code plots trace and correlation-vs-sigma comparisons.

Table1, Table2, Table3, and Table4 contain the source CSV/JSON inputs and render scripts for manuscript tables. Table1 calculates the spike inference precision, recall, F-score, accuracy, bias, and uncertainty summary from the Figure 4 matching artifacts before generating its CSV, TeX, and PDF. Table3 contains the GCaMP parameter JSON files also used by supp_figure6. The PowerShell render helpers regenerate TeX and PDF artifacts beneath output/Table1 through output/Table4 and require pdflatex; Python-backed helpers should be run from the pinned environment or passed the appropriate Python executable where supported.

## Python entry points

After pip install -e ., available console scripts include:

```
run_figure3_F_G_code
run_figure4_E_code
run_supp_figure8_code
run_supp_figure12_A_code
run_supp_figure13_code
run_supp_figure14_C_D_code
run_supp_figure15_F_G_code
```

The supplementary Python scripts accept --output-dir, --format, and --show where applicable. Temporary Matplotlib/cache directories created by these scripts are cleaned at the end of each run.

## Shared data and utilities

Shared_data contains common MATLAB data sources such as stopped-flow traces and the GCaMP database. Utilities contains shared MATLAB helpers, including AccessFitFiles_Janelia.m, GCaMPalias.m, and color utilities. SharedUtilities contains shared Python helpers, including trialwise_plots.py and project path resolution used by installed console scripts.